

CMPE2600 Lab 2 – Chem Plant – Fill Drain Cycle

Name: _____

Introduction

This project has multiple phases. All phases relate to designing and implementing a PLC program to control a chemical process. The actual plant uses 4 tanks: 2 chemical storage tanks, a mixing tank to heat and stir the two chemicals before dumping them to a holding tank. You will be programming the controller for a simulator that has a mixing tank and a reservoir representing the source tanks for the two chemicals and the holding tank.

System Description (effectively the Problem Statement)

The system has the following inputs and outputs:

- Start (digital input): momentary switch used to start the process initially and after it has been stopped and dumped. This is connected to Digital Input 11 on the PLC Trainer.
- Stop (digital input): momentary switch used to stop the process at any point! This is connected to Digital Input 9 on the PLC Trainer.
- Dump (digital input): momentary switch used to control dumping of the mixing tank AFTER a stop. This is connected to Digital Input 12 on the PLC Trainer.
- Mixing Tank Level (analog input): 4-20 mA signal from a differential pressure cell on the chemical plant. This signal is connected to Analog Input 3. The analog input reads a raw value in the range of your span for current in the range 4-20 mA. This requires some complex conversion to convert the raw value to an engineering value in the range 0 (for 4 mA) to 85cm (for 20 mA). Note that the pressure cell reading for zero may be slightly above 4 mA and a correction could be applied.
- Dump Valve (digital output): an open collector circuit that drives the dump valve solenoid at the bottom of the mixing tank. The dump valve is closed when energized. This is connected to Digital Output 10 on the PLC.
- Mixer (digital output): an open collector circuit that drives the agitator at the top of the mixing tank. This is connected to Digital Output 9 on the PLC.
- Pump (digital output): an open collector circuit that drives the pump feeding the mixing tank. Note that the pump must be on AND the fill valve must be open for the mixing tank to fill. The fill valve should be open whenever the pump is running. This is connected to Digital Output 8 on the PLC.
- Fill Valve (analog output): 4-20 mA signal driving a current to pressure (I to P) converter that controls an analog valve. The valve should receive 4 mA to be held in the closed position and 20 mA to be held in the open position. This requires some conversion to translate between the raw value (your span) and the engineering value (4-20 mA) to apply the appropriate value for closed. This is connected to Analog Output 1 on the PLC.

In addition to the operations described above, the system also needs to be able to 'prove' the turbine meter, so include the inputs and activities from the Proving the Turbine Meter document.

The process has 3 major modes: Normal, Reverse and Proving, as described below.

Initially the process is stopped and in the Waiting state. The tank is empty and the dump valve is open to keep it empty. At this point the system will respond to only 4 inputs: Start, Reverse, Prove and Emergency Stop.

If the Start button is pressed the system enters Normal mode which operates as follows:

- When the start button is pressed, the mixing tank fills to a preset level or 30cm of 'Chemical A'. This requires opening the fill valve and running the pump. The fill level value must be in a PLC counter that can be changed by your instructor prior to downloading the program.
- After the tank is filled to the preset level with 'Chemical A', the mixer runs for a preset duration 1 second for every 10 cm of fill. The preset duration must be in a PLC timer Set Value that can be changed by your instructor prior to downloading the program.
- After the mix time has expired, the mixing tank is dumped to the reservoir (representing the holding tank).
- If Stop is pressed during any of the previous steps, the process is stopped. The fill valve must be closed, the pump turned off, the mixer stopped, and the dump valve held closed.
- Pressing the Dump button from the Stop state causes the mixing tank to be dumped to the reservoir (representing the drain?). The Dump button is active only from the Stop state. After the mixing tank is empty, the process returns to the Waiting state.
- The proving activities may begin only if Prove is pressed before Start while in the Waiting state. Otherwise all of the inputs associated with the proving activities should be ignored. When proving is taking place the Start, Stop and Dump inputs should be ignored. You may wish to use Sequencer 1 for Normal plant operation and Sequencer 2 for the Proving activities.
- In Normal mode the 5 inputs associated with the Proving mode are to be ignored.

If Prove is pressed while in the Waiting state, then the Proving mode is entered where:

1. The proving activities from the Proving the Turbine Meter document are implemented.
2. The Start, Reverse and Dump, input will be ignored until the Waiting state is re-entered or Emergency Stop pressed which would allow for Dump.
3. Stop works anytime.

If Reverse is pressed while in the Waiting state, then the Reverse Mode is entered where:

1. Lab 6 implements Reverse.
2. The Start, Reverse and Dump, input will be ignored until the Waiting state is re-entered or Emergency Stop pressed respectively.
3. Stop works at any time.

Check off _____

